Chapter 14. Subsequence Conditions on Permutations

So Hard to Avoid

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14.1 Pattern avoidance

Motivation. Every length-3 pattern is avoided by c_n permutations; length 4 splits into three classes.

A lineup of children is good (each sees every shorter child ahead) iff there are no entries a, b, c, in this order, with $a < c < b$: 6723415 is good, 1423567 is not (4 blocks 1 from 2).

Definition 14.1 (132-pattern)

Entries a, b, c of a permutation, in this left-to-right order, with $a < c < b$ form a 132**-pattern** (they are ordered like $1, 3, 2$). Similarly for 123, 231, etc.

Definition 14.2 (132-avoiding)

A permutation with no 132-pattern is 132**-avoiding**.

Let $f(n)$ count 132-avoiding n-permutations ($f(0) = 1$). With n in position i, every entry left of n exceeds every entry right of it (else $x\,n\,y$ is a 132), so the sides are 132-avoiders of $\{n-i+1, \dots, n-1\}$ and $\{1, \dots, n-i\}$, and conversely. Hence

$$f(n) = \sum_{i=1}^{n} f(i-1)f(n-i) \qquad (n \geq 1). \tag{14.1}$$

Corollary 14.3

The number of 132-avoiding n-permutations is the Catalan number $c_n = \frac{1}{n+1}\binom{2n}{n}$.

Proof. (14.1) with $f(0) = 1$ is the Catalan recurrence $c_n = \sum_{i=1}^{n} c_{i-1}c_{n-i}$ (from (8.6), §8.1.2.1); induct. ∎

Definition 14.4 (Pattern)

Let p be an n-permutation and $q = q_1 \cdots q_k$ a k-permutation, $k \leq n$. Entries $a_1, \dots, a_k$ of p (left to right) form a q**-pattern** if $a_i < a_j \iff q_i < q_j$.

Definition 14.5 (Avoidance)

p is q**-avoiding** if no k entries of p form a q-pattern. $S_n(q)$ (or $\mathrm{Av}_n(q)$) denotes the number of q-avoiding n-permutations.

Symmetries. The **reverse** $p_n \cdots p_1$ and the **complement** $\bar{p} = (n+1-p_1)\cdots(n+1-p_n)$ biject q-avoiders onto avoiders of the reversed/complemented pattern, so

$$S_n(132) = S_n(231) = S_n(312) = S_n(213), \qquad S_n(123) = S_n(321).$$

A **left-to-right minimum** is an entry smaller than all before it; these form a decreasing subsequence from p_1 to 1.

Lemma 14.6

$S_n(123) = S_n(132)$ for all $n \geq 1$.

Proof. A bijection from 123- to 132-avoiders fixing the left-to-right minima (values and positions).

Step 1 (constraints). Let $m_1 > \cdots > m_r$ be the minima at positions $1 = i_1 < \cdots < i_r$. The other positions are *slots*; a slot between i_j and i_{j+1} has *threshold* m_j. A filling keeps exactly these minima iff each slot value exceeds its threshold. Thresholds decrease.

Step 2 (the map f). For a 123-avoider p, fill slots left to right with the smallest unused value above the threshold. At the s-th slot (threshold m), the first s slots of p hold s values $> m$ and only $s-1$ are used, so this never gets stuck.

Step 3 ($f(p)$ avoids 132). Non-minima x before y, both above a minimum m_j to their left, satisfy $x < y$ (y was available when x was placed). A 132-pattern a, b, c may take $a = m_j$ (the last minimum at or before a); then $b, c > m_j$ are non-minima to its right, so $b < c$, a contradiction.

Step 4 (the inverse g). For a 132-avoider q, fill the slots with the non-minima in *decreasing* order; this respects thresholds (the first s slots of q hold s values above the s-th threshold). $g(q)$ is a union of two decreasing sequences, so avoids 123.

Step 5 (g and f are inverse). In a 123-avoider the non-minima decrease (else the minimum before x, x, y is a 123), so $g(f(p)) = p$. A 132-avoider q is the greedy filling: if a slot (threshold m_j) holds x while a smaller eligible $y > m_j$ comes later, m_j, x, y is a 132. So $f(g(q)) = q$. ■

E.g. $p = 465132$ has minima $4, 1$ and $f(p) = 456123$.

Theorem 14.7

For every pattern q of length 3 and all $n \geq 1$, $S_n(q) = c_n = \frac{1}{n+1}\binom{2n}{n}$.

Proof. Symmetries, Lemma 14.6 and Corollary 14.3. ■

Length 4 leaves three classes 1342, 1234, 1324:

$$\begin{array}{ll} S_n(1342): & 1, 2, 6, 23, 103, 512, 2740, 15485 \\ S_n(1234): & 1, 2, 6, 23, 103, 513, 2761, 15767 \\ S_n(1324): & 1, 2, 6, 23, 103, 513, 2762, 15793 \end{array}$$

($n \leq 8$). In fact $S_n(1342) < S_n(1234) < S_n(1324)$ for $n \geq 7$; we prove the second inequality.

Theorem 14.8

$S_n(1234) < S_n(1324)$ for all $n \geq 7$.

A **right-to-left maximum** exceeds all entries to its right. Proof after Lemma 14.11.

Definition 14.9 (Class)

Permutations x, y are in the same **class** if they have the same left-to-right minima at the same positions and the same right-to-left maxima at the same positions.

E.g. 51234, 51324 share a class; 24315, 24135 do not.

Constraints of a class [Supplement]. Each remaining position (slot) s has a lower bound $l(s)$ (nearest minimum to its left) and upper bound $u(s)$ (nearest maximum to its right); a filling lies in the class iff $l(s) <$ value $< u(s)$ for all slots. l, u are non-increasing.

Lemma 14.10

Every nonempty class contains exactly one 1234-avoiding permutation.

Proof.

Step 1 (existence). Fill the slots in decreasing order. [Supplement] In any member, the first s slots hold s values $> l(s)$ and the last $N-s+1$ slots values $< u(s)$, so the s-th largest value lies in $(l(s), u(s))$. The result is a union of three decreasing sequences, so by pigeon-hole has no 1234.

Step 2 (uniqueness). In a 1234-avoider, slot entries $a < b$ in this order with the minimum before a and the maximum after b would form 1234; so slot entries decrease, as in Step 1. ■

An **inversion** of p is a pair $i < j$ with $p_i > p_j$.

Lemma 14.11

Every nonempty class contains at least one 1324-avoiding permutation.

Proof.

Step 1 (good patterns). In a 1324-pattern $a\,c\,b\,d$, replace a by the nearest minimum at or before it and d by the nearest maximum at or after it: a **good pattern**. Its middle entries c, b are slot entries (minima can only play "1", maxima "4").

Step 2 (swapping). Exchange c and b: at its new place $b > a \geq l$ and $b < c < u$; $c < d \leq u$ and $c > b > l$. So the class is kept, and the number of inversions drops.

Step 3 (termination). After at most $\binom{n}{2}$ swaps no good pattern, hence no 1324, remains. ■

Proof of Theorem 14.8. Lemmas 14.10, 14.11 give $S_n(1234) \leq S_n(1324)$. Write $a_1{*}a_2{*}{*}b_1$ for the class with minima a_1, a_2 at positions $1, 3$ and maximum b_1 at position 6. For $n = 7$, $3{*}1{*}7{*}5$ contains the two 1324-avoiders 3612745, 3416725; for $n > 7$ use $n(n-1)\cdots 8\,3{*}1{*}7{*}5$. ■

Stanley–Wilf. $S_n(q) \leq c_q^n$ for every q (Marcus–Tardos, 2004; cited). For monotone patterns:

Theorem 14.12

For $k \leq n$, $S_n(12\cdots k) \leq (k-1)^{2n}$.

Proof. The **rank** of x is the length of the longest increasing subsequence ending at x; equal ranks form decreasing subsequences. A $12\cdots k$-avoider has ranks in $[k-1]$, and is determined by assigning values ($(k-1)^n$ ways) and positions ($(k-1)^n$ ways) to the $k-1$ decreasing subsequences. ■

Compare $S_n(123) = c_n < 4^n$.

Definition 14.13 (Direct sum)

For $p \in S_a$, $q \in S_b$, the **direct sum** $p \oplus q \in S_{a+b}$ is p followed by q with every entry increased by a.

Example 14.14

$132 \oplus 2431 = 1325764$.

Theorem 14.15

If $S_n(q_1 \oplus 1) \leq c_1^n$ and $S_n(1 \oplus q_2) \leq c_2^n$ for all n, then

$$S_n(q_1 \oplus 1 \oplus q_2) \leq \left(\sqrt{c_1} + \sqrt{c_2}\right)^{2n}.$$

Example 14.16

$S_n(2134) = S_n(1234) < 9^n$ (Wilf-equivalence cited; bound by Theorem 14.12), likewise $S_n(1243) < 9^n$. With $q_1 = 213$, $q_2 = 132$: $S_n(213 \oplus 1 \oplus 132) \le (3+3)^{2n} = 36^n$.

Proof of Theorem 14.15. Let p avoid $q = q_1 \oplus 1 \oplus q_2$; colour an entry red if it ends some $(q_1 \oplus 1)$-pattern, blue otherwise.

Step 1. Red entries avoid $1 \oplus q_2$: the first entry x of a red copy ends a $q_1 \oplus 1$ whose other entries are left of and below x, completing a q.

Step 2. Blue entries avoid $q_1 \oplus 1$ (its last entry would be red).

Step 3 (counting). With k blue entries, p is fixed by the blue values, blue positions and the two patterns: $\le \binom{n}{k}^2 c_1^k c_2^{n-k}$. So

$$S_n(q) \le \sum_{k=0}^{n} \left(\binom{n}{k} \sqrt{c_1}^{\,k} \sqrt{c_2}^{\,n-k} \right)^2 \le \left(\sum_{k=0}^{n} \binom{n}{k} \sqrt{c_1}^{\,k} \sqrt{c_2}^{\,n-k} \right)^2 = \left(\sqrt{c_1} + \sqrt{c_2} \right)^{2n}.$$

■

14.2 Stack sortable permutations

Motivation. One stack sorts exactly the 231-avoiders; $s(LnR) = s(L)s(R)n$ reveals a descent symmetry.

Stack sorting. Read p left to right; push an entry if it is smaller than the top, otherwise pop the top to the output until it can be pushed; empty the stack at the end. The output is $s(p)$; p is **stack sortable** if $s(p) = 12 \cdots n$.

Example 14.17

$p = 2413$: push 2; 4 pops 2, pushed; push 1; 3 pops 1, pushed on 4; pop 3, 4. So $s(2413) = 2134$: not stack sortable.

Proposition 14.18

Let $a < b$ be entries of p.
(1) If a precedes b in p, then a precedes b in $s(p)$.
(2) If b precedes a and no entry $c > b$ lies between them, then a precedes b in $s(p)$.
(3) If b precedes a and some $c > b$ lies between them (so b, c, a form a 231-pattern), then b precedes a in $s(p)$.

Proof.

(1) b cannot go on top of a, so a leaves first.

(2) An entry leaves only when a larger one arrives or at the end; so b is still stacked when a goes above it. [Supplement] (The book's claim that the entries between are decreasing is not needed.)

(3) b leaves before c is pushed, which is before a arrives. ■

Theorem 14.19

p is stack sortable iff it avoids 231.

Proof. A 231 $b\,c\,a$ keeps b before a (part 3); otherwise every pair falls under (1) or (2). ■

So c_n n-permutations are stack sortable. p is **two-stack sortable** if $s(s(p))$ is the identity; this is not closed under patterns:

Example 14.20

$s(35241) = 32145$ and $s(32145) = 12345$, so 35241 is two-stack sortable. But its subsequence 3241 has $s(3241) = 2314$ and $s(2314) = 2134$.

Theorem 14.21 (West)

p is two-stack sortable iff it contains no 2341-pattern and no 3241-pattern except as part of a 35241-pattern.

Proof.

Step 1 (a 2341 obstructs). For $b\,c\,d\,a$ ($a < b < c < d$), b precedes c (part 1) and both precede a (part 3, via d) in $s(p)$: a 231.

Step 2 (a bad 3241 obstructs). For $y\,x\,z\,w$ ($w < x < y < z$) not in a 35241, x, y precede w in $s(p)$ (via z). If nothing between y, x exceeds y, x precedes y (part 2) and x, y, w is a 231. Else some such t has $t < z$ (no 35241), so $y\,t\,z\,w$ is a 2341 (Step 1).

Step 3 (sufficiency). Let $f\,g\,e$ be a 231 in $s(p)$ ($e < f < g$). By Proposition 14.18, f, g precede e in p with some $h > g$ between g and e.

(a) If f precedes g in p, then $f\,g\,h\,e$ is a 2341.

(b) If g precedes f: nothing between them exceeds g (part 3), so h is between f and e and $g\,f\,h\,e$ is a 3241, not inside a 35241 (which needs an entry $> h$ between g and f). ■

Two-stack sortable n-permutations number $\frac{2}{(n+1)(2n+1)}\binom{3n}{n}$ (cited). p is **t-stack sortable** if $s^t(p)$ is the identity.

Lemma 14.22

If $p = LnR$, then $s(p) = s(L)\,s(R)\,n$.

Proof. L leaves before n enters, giving $s(L)$; R passes over n, giving $s(R)$; n leaves last. ■

Corollary 14.23

Every n-permutation is $(n-1)$-stack sortable.

Proof. Induction on n: $s(\sigma n) = s(\sigma)n$, so $s^{n-1}(p) = s^{n-2}(s(L)s(R))\,n$, with $s(L)s(R)$ a permutation of $[n-1]$. ■

Corollary 14.24

$s^t(p)$ ends with $(n-t+1)(n-t+2)\cdots n$.

Proof. Induction on t [Supplement]: if $s^{t-1}(p) = \sigma\,(n-t+2)\cdots n$, Lemma 14.22 gives $s^t(p) = s(\sigma)\,(n-t+2)\cdots n$, and $s(\sigma)$ ends with $\max\sigma = n-t+1$. ■

Decreasing binary trees. $T(p)$ has root n, left child $\max L$ and right child $\max R$ (an only child is still left or right), with $T(L)$, $T(R)$ below. In-order reading recovers p.

Example 14.25

For $p = 263498175$, $T(p)$ is drawn below.

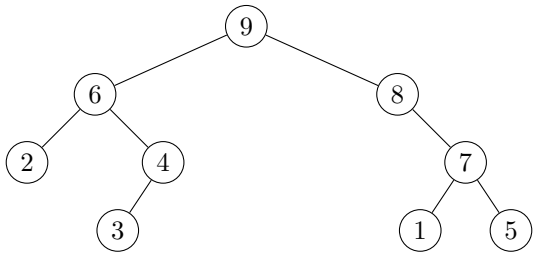


Figure. $T(263498175)$*: in order* 263498175*, postorder* 234615789 $= s(p)$.

Example 14.26

Postorder (left, right, root) of the tree above gives $234615789 = s(263498175)$.

Proposition 14.27

$T(p)$ read in postorder is $s(p)$.

Proof. Induction: postorder of $T(LnR)$ is $s(L)s(R)n = s(p)$ (Lemma 14.22). ■

i is a **descent** if $p_i > p_{i+1}$; $d(p)$ counts them. By complementation, k and $n-1-k$ descents are equally frequent overall. [Supplement] $p_i > p_{i+1}$ iff p_i has a right child (its in-order successor is then the leftmost vertex of its right subtree, else an ancestor), so $d(p)$ is the number of right edges of $T(p)$.

Let $W_t(n,k)$ count t-stack sortable n-permutations with k descents. Reverse and complement do not preserve t-stack sortability (213, 132), yet $W_t(n,k) = W_t(n,n-1-k)$.

Definition 14.28 (The map f)

Define f on permutations (of any finite set of integers) by $f(1) = 1$ and, with n the maximum: $f(LnR) = f(L)\,n\,f(R)$ if L, R are nonempty; $f(Ln) = n\,f(L)$; $f(nR) = f(R)\,n$.

Example 14.29

$f(123) = 3\,f(12) = 3\,2\,f(1) = 321$, so $f(4123) = f(123)\,4 = 3214$.

Example 14.30

$f(1423) = f(1)\,4\,f(23) = 1432$.

Example 14.31

$f(412395867) = f(4123)\,9\,f(5867) = 3214\,9\,5876$.

Proposition 14.32

$d(p) + d(f(p)) = n - 1$ for every n-permutation p.

Proof. Induction on n; $n = 1$ is trivial.

Step 1 ($p = LnR$, $L, R \neq \emptyset$, n in position i). $d(p) = d(L)+d(R)+1$, $d(f(p)) = d(f(L))+d(f(R))+1$, and $d(L) + d(f(L)) = i - 2$, $d(R) + d(f(R)) = n - i - 1$: total $n - 1$.

Step 2 ($p = Ln$). $d(L) + d(f(L)) + 1 = (n - 2) + 1$.

Step 3 ($p = nR$). $1 + d(R) + d(f(R)) = 1 + (n - 2)$. ■

Lemma 14.33

$s(p) = s(f(p))$ for every permutation p.

Proof. Induction, Lemma 14.22, and $s(L) = s(f(L))$, $s(R) = s(f(R))$:

(1) $p = LnR$: $s(p) = s(L)s(R)n = s(f(L))s(f(R))n = s(f(L)nf(R)) = s(f(p))$.

(2) $p = nR$: $s(p) = s(R)n = s(f(R))n = s(f(R)n) = s(f(p))$.

(3) $p = Ln$: $s(p) = s(L)n = s(f(L))n = s(nf(L)) = s(f(p))$. ■

Corollary 14.34

p is t-stack sortable iff $f(p)$ is.

Proof. Both mean $s(p) = s(f(p))$ is $(t - 1)$-stack sortable. ■

Theorem 14.35

$W_t(n, k) = W_t(n, n - 1 - k)$ for all n, k, t.

Proof. f is a bijection (on $T(p)$ it flips every only child, an involution). By Corollary 14.34 and Proposition 14.32 it maps t-stack sortable permutations with k descents onto those with $n - 1 - k$. ■

Tree view. f flips every only child and keeps two-child vertices, swapping left and right edge counts (Proposition 14.32); postorder ignores the side of an only child (Lemma 14.33), at every subtree.

Example 14.36

$p = 356124$ and $f(p) = 536421$ both give 351246 in postorder, also on each subtree.